

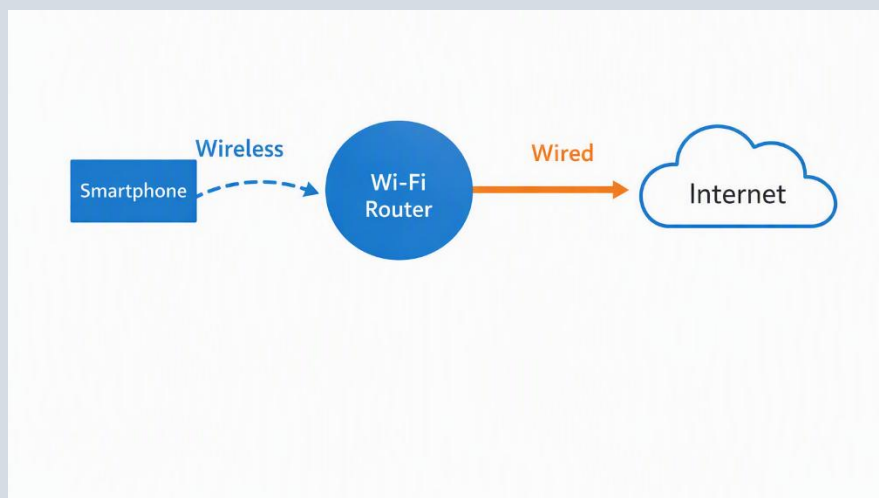
Wireless networking

1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.



Wi-Fi Standard	Key Difference
802.11n (Wi-Fi 4)	Works on 2.4 GHz and 5 GHz bands with speeds up to 600 Mbps.
802.11ac (Wi-Fi 5)	Works only on the 5 GHz band and offers speeds up to 3.5 Gbps (faster than 802.11n).
802.11ax (Wi-Fi 6)	Works on 2.4 GHz and 5 GHz bands (Wi-Fi 6E also supports 6 GHz) and provides speeds up to 9.6 Gbps with better performance for many connected devices.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.
Hint: Consider both signal strength and security when choosing.

SSID (Wi-Fi Name)	Signal Strength	Security Type
laxminarayan	Medium (Good)	WPA2-Personal
patelhensie	Strong (Excellent)	WPA2-Personal
Redmi Note 13 Pro 5G	Weak (Fair)	WPA2-Personal

Which network would I choose and why?

I would choose patelhensie because:

- It has a Strong (Excellent) signal, which provides faster and more stable internet.
- It uses WPA2-Personal security, which is secure for home Wi-Fi networks.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID Hiding (Hidden Wi-Fi Network):

SSID hiding means the Wi-Fi network name (SSID) is not shown in the list of available Wi-Fi networks. Users must manually enter the network name and password to connect.

Advantage:

- It makes the Wi-Fi network less visible to casual users, reducing unwanted connection attempts.

Disadvantage:

- It does not provide real security because the hidden SSID can still be detected using network scanning tools. It can also make it less convenient for your own devices to connect.